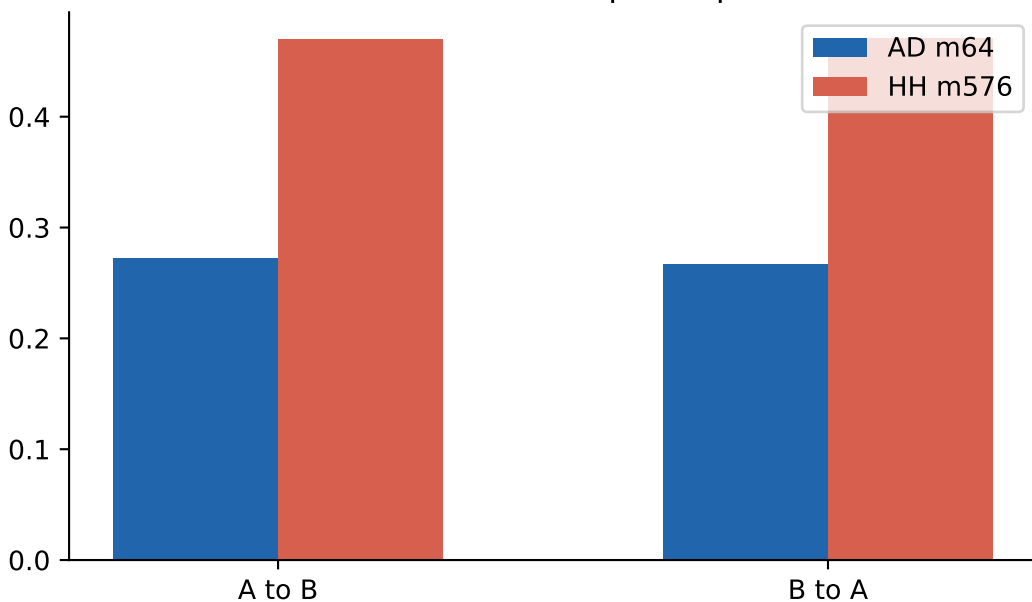
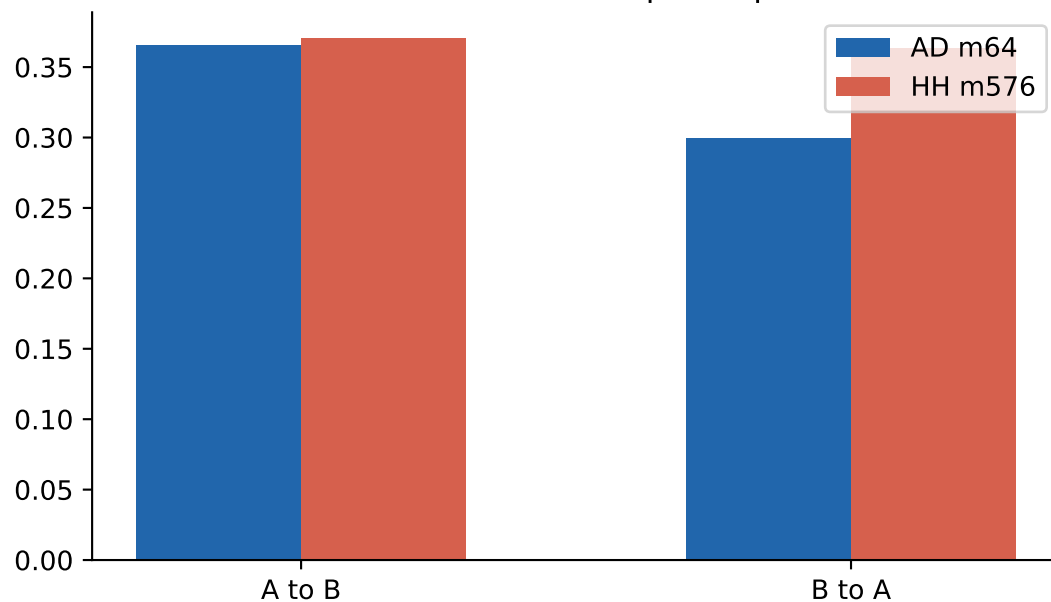


Direct product training; exact 73,729 trainable parameters per head

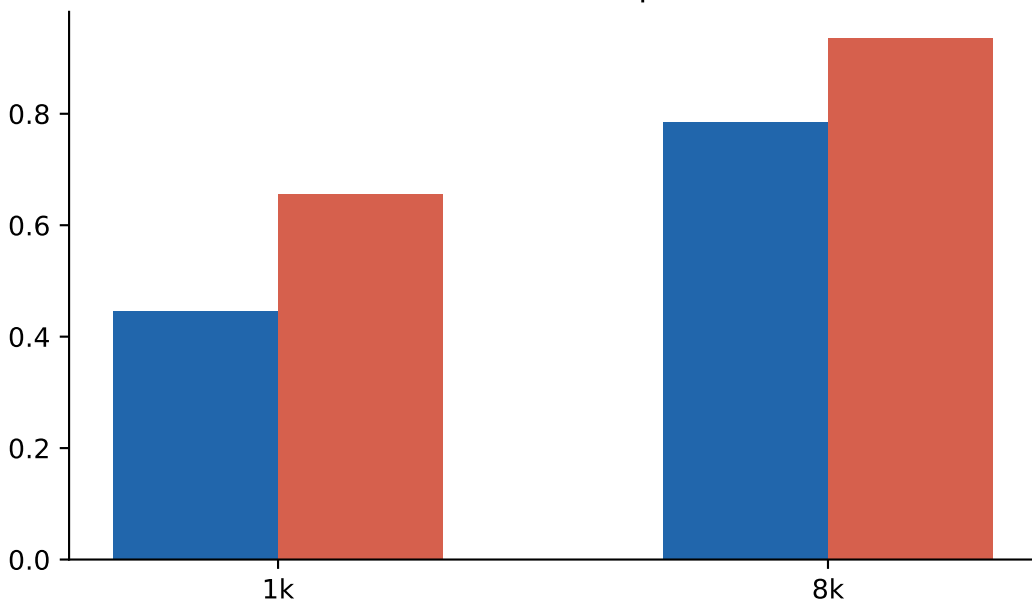
Raw l risk: heldout empirical products



Raw L2 risk: heldout empirical products



Causal attention output NMSE



Frozen model PPL: 24/336 heads replaced

